MEHARAJ SAMI SIAM

Applied AI Engineer — Agentic AI, LLM, RAG & MCP

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PROFESSIONAL SUMMARY

Applied AI Engineer building production, agentic AI and **Large Language Model (LLM)** applications — three-tier LLM routing, **Retrieval-Augmented Generation (RAG)** pipelines, **Model Context Protocol (MCP)** integrations, and LLM fine-tuning on **AWS Bedrock (Anthropic Claude)**. Strong full-stack (Django, Next.js, Vue) and REST API / SDK engineering foundation, with hands-on machine learning, prompt engineering, and scalable cloud deployment (AWS). Has delivered **13+ projects end-to-end**, from architecture and design through production deployment, including a multi-tenant AI platform hardened to enterprise security standards and **6 production SDKs**.

CORE COMPETENCIES

Agentic AI & MCP | Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG) | LLM Fine-Tuning | Prompt Engineering | Machine Learning | Generative AI | AWS Bedrock (Claude) | REST API & SDK Design | API Integration | Multi-Tenant Architecture | Full-Stack (Django, Next.js, Vue) | Application Security | Cloud & DevOps (AWS, CI/CD)

PROFESSIONAL EXPERIENCE

Software Engineer — WebAlive Jan 2026 – Present

- Built **Cortex AI**, a multi-tenant AI agent platform: a three-tier **Large Language Model (LLM)** routing engine (confidence, action, and knowledge tiers) on AWS Bedrock (Anthropic Claude), a **Retrieval-Augmented Generation (RAG)** pipeline with vector search, a Django / Django REST Framework backend and Vue dashboard, and a plugin protocol — delivered in a focused six-week sprint and deployed on **AWS ECS (Fargate)**.
- Led a comprehensive **security-hardening** initiative: Argon2 password hashing, split-key JWT with revocation, SSRF and CORS allowlisting, HttpOnly cookies with CSRF protection, role-based access control (RBAC), Content-Security-Policy, PII redaction and retention controls, per-tenant rate limiting, IAM-scoped model access, and **RAG prompt-injection mitigation**.
- Built **Store Migration AI**, a **Model Context Protocol (MCP)**-based Shopify-to-WebCommander migration pipeline with machine-learning-driven API matching, and drove API standardization for AI-agent and MCP consumption.
- Lead developer on a **clinical conversational AI portal** (Next.js): session and survey lifecycle, a form scenario engine, and a two-tier chatbot lifecycle; reviewed the RAG ingestion pipeline for a clinical-genetics knowledge assistant.

Software Engineer — Bit Mascot (Pvt.) Ltd. Jun 2024 – Dec 2025

- Architected and shipped **6 production REST API SDKs** (Python, PHP, C# / .NET) for subscription-commerce, events, and e-commerce platforms — 25+ entity modules each with full CRUD, OAuth 2.0 authentication, thread-safe token refresh, pagination, and API versioning.
- Built **E.L.I Bot**, an enterprise customer-support chatbot, with an **LLM fine-tuning** pipeline, a custom text-chunking algorithm, training UX/UI, and a document-to-training-data flow.
- Delivered core backend APIs (PHP) for a subscription-commerce platform and **tech-led a C# / .NET SDK**, setting the architecture for HTTP-client management, dual authentication, and API versioning, and reviewing the full codebase.

Founder & Lead Developer — Medview 2022 – 2024

- **Government-funded (Bangladesh ICT Division)** medicine-management platform with OCR prescription reading — led the team and technical strategy from concept through build.

Co-Founder & CTO — Project Durjo 2022 – 2024

- **Government-funded (Bangladesh ICT Division)** disaster-management robotics system — owned the technical architecture and engineering direction.

PROJECTS

Dorpon — Agent-Ready Eyewear Commerce
Live demo: dorpon-eyewear.netlify.app | Code: github.com/Meharaj002

- Full-stack premium eyewear store (Next.js, Django / DRF) with browser-based **virtual try-on** and an AI shopping concierge on AWS Bedrock (Anthropic Claude).
- Exposed the store to AI agents through a stateless **Model Context Protocol (MCP)** server and a **Universal Commerce Protocol (UCP)** REST layer (discovery manifest, JSON-Schema actions, RFC 9457 errors), aligned with the Agentic Commerce Protocol — one shared tool registry powering the web app, MCP, and REST. Deployed on free-tier cloud infrastructure (Render, Netlify, Neon, Supabase).

TECHNICAL SKILLS

AI & Machine Learning: Agentic AI, Model Context Protocol (MCP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) — chunking, embeddings, vector databases, LLM fine-tuning, prompt engineering, Generative AI, Natural Language Processing (NLP), AWS Bedrock (Anthropic Claude), prompt-injection mitigation; PyTorch, TensorFlow, Keras
Languages: Python, PHP, JavaScript, TypeScript, C# / .NET, Java, C/C++, SQL
Frameworks & Libraries: Django, Django REST Framework (DRF), Next.js, Vue.js, React, Vitest
Databases & Caching: PostgreSQL, Redis, DynamoDB, MySQL, SQLite, vector databases
Cloud & DevOps: AWS (ECS / Fargate, Bedrock, IAM), CI/CD, Git, GitHub, Bitbucket, Git LFS
Security: OAuth 2.0, JWT, Argon2, CSRF / CORS / SSRF protection, RBAC, Content-Security-Policy, rate limiting, PII handling
Practices & Tools: REST API design, SDK design, API integration, microservices, multi-tenant architecture, Agile, Jira, Confluence

EDUCATION & CREDENTIALS

- **B.Sc. in Computer Science** — BRAC University (2020 – 2024)
- **Certifications: Anthropic AI Certifications** — Anthropic; Python Programming · Python Data Structures — University of Michigan; Social Media Marketing — Meta
- **Award:** Runner-Up, Project Showcase — 5th International Tech Carnival, Bangladesh ICT Division